

Organizational

Natural Language Processing for the Digital Humanities

Winter semester 2026/27

Martina Scholger and Lucija Brozić
Department of Digital Humanities

Contact

Martina Scholger

- martina.scholger@uni-graz.at
- Senior Scientist, Chair of the „Digital Humanities“ Curriculum Commission
- Focus areas: Text and data modeling, Text Encoding Initiative, LLMs, Digital Scholarly Editing, Text and data mining



Lucija Brozić

- lucija.brozic@uni-graz.at
- post-doc at DDH Graz
- Focus areas: Natural Language Processing, Machine Learning, corpus building and annotation, evaluation



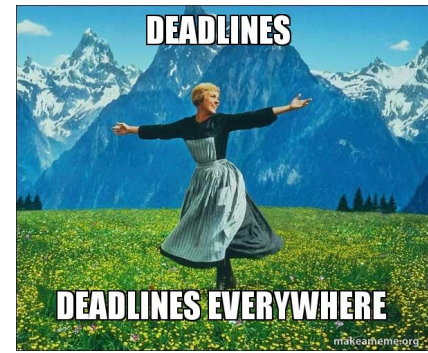
Course content

- learn about methods from the field of Natural Language Processing
- how and why these methods can be used to answer RQ from the Digital Humanities
- apply NLP methods to real-world humanities data in history, literature, art, and cultural studies
- devise NLP projects, create open-source models and achieve reproducible results (annotation, training, evaluation)
- assess the optimal NLP methodologies for your own use cases

About the course

- **Course materials** can be found on the Moodle learning platform (<http://moodle.uni-graz.at/>)
- **Software:** Python: Anaconda and Google Colab (GPU), your preferred Python environment
- **Grading:** Homework (60%), exam (40%, January 27th) - both parts must be positive
- **Regular attendance** (no more than **three** missed units)

Deadlines



- **All deadlines are strict.**
- If there are serious reasons, it is possible to extend the submission of examination-relevant work.
 - Good reasons: Illness, childcare, or comprehensible reasons communicated promptly to the lecturer.
- If a deadline is missed
 - Without communication with the lecturer, missed submissions will be assessed negatively.
- Plagiarism is not tolerated

Use of generative AI

The use of tools such as OpenAI's ChatGPT, OpenAI's GPT API, Bing, Bard etc. is permitted. However, explicit reference must be made to their use. All learning objectives defined in the course must nevertheless be achieved by the students themselves and may be checked in a submission discussion or using a comparable method. Please document the use of text generators in a similar way to a bibliographic reference.

Include the following:

- Title/purpose of the tool used,
- Technology and version of the tool used
- Date
- URL to the conversation

An example: GV has summarized Colavizza et al. 2022 with the support of GPT4. He adds the following information to the paragraph based on this:

Summary of Colavizza et al. 2022 with GPT-4, ChatGPT May 24 Version.
09.06.2023.

<https://chat.openai.com/share/525b50a6-c222-4d8c-b8e1-890292da4c96>.

Preliminary schedule

1.	Organizational, Introduction to NLP, GIT
2.	Python Revision
3.	Preparing data, pre-processing, OCR
4.	Preparing data, pre-processing, OCR
5.	Machine Learning
6.	Supervised ML methods: classification and prediction

Preliminary schedule

7.	Sentiment analysis
8.	Sentiment analysis
9.	Sentiment analysis
10.	NER
11.	Topic Modeling
12.	Topic Modeling
13.	Topic modeling
14.	Exam

Department of Digital Humanities

- Applied research in the field of information processing in the humanities
- Main focus:
 - Theory of information modeling
 - Text technology
 - Digital Edition
 - Digital Museology
- Practice-oriented implementation of the research subject in cooperation projects, research and teaching
- <https://digital-humanities.uni-graz.at/en/>

Studying Digital Humanities

Master's degree program

<https://digital-humanities.uni-graz.at/en/study-service/master-of-arts-in-digital-humanities/>

Information modelling certificate

<https://digital-humanities.uni-graz.at/en/study-service/module-and-certificate/>

Round of introductions

- Name, pronouns, interests
- Domain of origin
- Previous knowledge in Digital Humanities, Python, digital scholarly editing, XML/TEI, Machine Learning?
- Which languages are you able to read?

